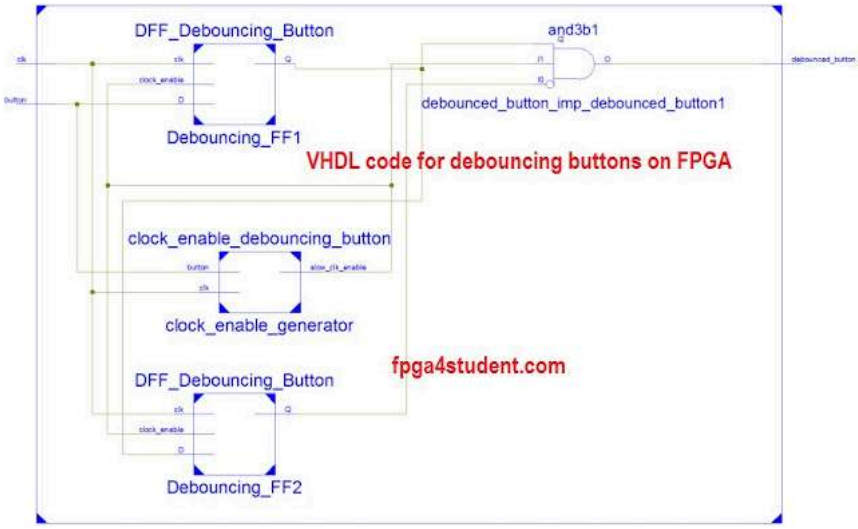
VHDL code for debouncing buttons on FPGA

When pressing buttons on FPGA, there are unpredictable bounces that are unwanted. This VHDL code is to debounce buttons on FPGA by only generating a single pulse with a period of the input clock when the button on FPGA is pressed, held long enough, and released. Last time, I presented a simple Verilog code for debouncing buttons on FPGA.

This VHDL project is to present a VHDL code for debouncing buttons on FPGA. Full VHDL code and testbench are provided.



VHDL code for debouncing buttons on FPGA:

```
--fpga4student.com: FPGA projects, Verilog projects, VHDL projects
-- VHDL project: VHDL code for debouncing buttons on FPGA
-- Generate Slow clock enable for debouncing buttons
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;
USE IEEE.STD_LOGIC_UNSIGNED.ALL;
entity clock_enable_debouncing_button is
port(
 clk: in std_logic; -- input clock on FPGA 100Mhz
                          -- Change counter threshold accordingly
 slow_clk_enable: out std_logic
);
end clock_enable_debouncing_button;
architecture Behavioral of clock_enable_debouncing_button is
signal counter: std_logic_vector(27 downto 0):=(others => '0');
begin
process(clk)
begin
if(rising_edge(clk)) then
  counter <= counter + x"0000001";
  if(counter>=x"003D08F") then -- reduce this number for simulation
   counter <=  (others => '0');
  end if;
 end if;
end process;
 slow_clk_enable <= '1' when counter=x"003D08F" else '0'; -- reduce this num
end Behavioral;
--fpga4student.com: FPGA projects, Verilog projects, VHDL projects
-- VHDL project: VHDL code for debouncing buttons on FPGA
-- VHDL D-flip-flop with clock enable signal for debouncing buttons
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;
entity DFF_Debouncing_Button is
```

```
port(
 clk: in std_logic;
 clock_enable: in std_logic;
 D: in std_logic;
 Q: out std_logic:='0'
);
end DFF_Debouncing_Button;
architecture Behavioral of DFF_Debouncing_Button is
begin
process(clk)
begin
 if(rising_edge(clk)) then
  if(clock_enable='1') then
   Q <= D;
  end if;
 end if;
end process;
end Behavioral;
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-- VHDL project: VHDL code for debouncing buttons on FPGA
-- VHDL code for button debouncing on FPGA
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;
entity Debouncing_Button_VHDL is
port(
 button: in std_logic;
 clk: in std_logic;
 debounced_button: out std_logic
);
end Debouncing_Button_VHDL;
architecture Behavioral of Debouncing_Button_VHDL is
signal slow_clk_enable: std_logic;
signal Q1,Q2,Q2_bar,Q0: std_logic;
begin

clock_enable_generator: entity work.clock_enable_debouncing_button PORT MAP
      ( clk => clk,
        slow_clk_enable => slow_clk_enable
      );

Debouncing_FF0: entity work.DFF_Debouncing_Button PORT MAP
      ( clk => clk,
        clock_enable => slow_clk_enable,
        D => button,
        Q => Q0
      );


Debouncing_FF1: entity work.DFF_Debouncing_Button PORT MAP
      ( clk => clk,
        clock_enable => slow_clk_enable,
        D => Q0,
        Q => Q1
      );
Debouncing_FF2: entity work.DFF_Debouncing_Button PORT MAP
      ( clk => clk,
        clock_enable => slow_clk_enable,
        D => Q1,
        Q => Q2
      );
 Q2_bar <= not Q2;
 debounced_button <= Q1 and Q2_bar;
end Behavioral;
```
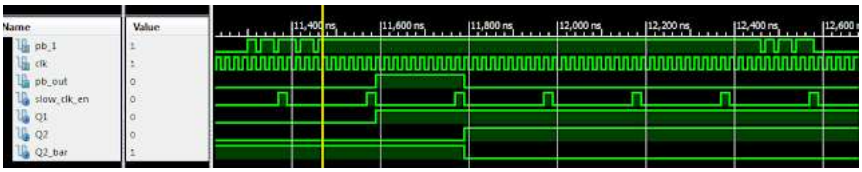

Testbench VHDL code for debouncing buttons on FPGA:

```
LIBRARY ieee;
USE ieee.std_logic_1164.ALL;
--fpga4student.com: FPGA projects, Verilog projects, VHDL projects
-- VHDL code for button debouncing on FPGA
-- Testbench VHDL code for button debouncing
ENTITY tb_debouncing_button_VHDL IS
END tb_debouncing_button_VHDL;

ARCHITECTURE behavior OF tb_debouncing_button_VHDL IS
    -- Component Declaration for VHDL code for button debouncing
    COMPONENT Debouncing_Button_VHDL
    PORT(
         button : IN  std_logic;
         clk : IN  std_logic;
         debounced_button : OUT  std_logic
        );
    END COMPONENT;
   signal button : std_logic := '0';
   signal clk : std_logic := '0';
   signal debounced_button : std_logic;
   constant clk_period : time := 10 ns;
BEGIN
 -- Instantiate VHDL code for button debouncing
   uut: Debouncing_Button_VHDL PORT MAP (
          button => button,
          clk => clk,
          debounced_button => debounced_button
        );
   clk_process :process
   begin
  clk <= '0';
  wait for clk_period;
  clk <= '1';
  wait for clk_period;
   end process;
   -- Stimulus process
   stim_proc: process
   begin
      -- hold reset state for 100 ns.
      wait for 100 ns;

      wait for clk_period*10;
  button <= '0';
  wait for 10 ns;
  button <= '1';
  wait for 20 ns;
  button <= '0';
  wait for 10 ns;
  button <= '1';
  wait for 30 ns;
  button <= '0';
  wait for 10 ns;
  button <= '1';
  wait for 40 ns;
  button <= '0';
  wait for 10 ns;
  button <= '1';
  wait for 30 ns;
  button <= '0';
  wait for 10 ns;
  button <= '1';
  wait for 1000 ns;
  button <= '0';
```

```
  wait for 10 ns;
  button <= '1';
  wait for 20 ns;
  button <= '0';
  wait for 10 ns;
  button <= '1';
  wait for 30 ns;
  button <= '0';
      wait;
   end process;

END;
```

Simulation Waveform for the VHDL code:



Only a single pulse is created as shown in the simulation waveform above when the input button is pressed, held long enough, and released.

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